

Gamma activity modulated by naming of ambiguous and unambiguous images: Intracranial recording



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HIGHLIGHTS

- Ambiguous and unambiguous images differentially elicited cortical gamma-augmentation.
- Ambiguous image-specific activation involved frontal and parietal areas at >400 ms.
- We failed to replicate rapid top-down guess processing exerted by the orbitofrontal cortex.

ABSTRACT

Objective: Humans sometimes need to recognize objects based on vague and ambiguous silhouettes. Recognition of such images may require an intuitive guess. We determined the spatial-temporal characteristics of intracranially-recorded gamma activity (at 50–120 Hz) augmented differentially by naming of ambiguous and unambiguous images.

Methods: We studied 10 patients who underwent epilepsy surgery. Ambiguous and unambiguous images were presented during extraoperative electrocorticography recording, and patients were instructed to overtly name the object as it is first perceived.

Results: Both naming tasks were commonly associated with gamma-augmentation sequentially involving the occipital and occipital-temporal regions, bilaterally, within 200 ms after the onset of image presentation. Naming of ambiguous images elicited gamma-augmentation specifically involving portions of the inferior-frontal, orbitofrontal, and inferior-parietal regions at 400 ms and after. Unambiguous images were associated with more intense gamma-augmentation in portions of the occipital and occipital-temporal regions.

Conclusions: Frontal-parietal gamma-augmentation specific to ambiguous images may reflect the additional cortical processing involved in exerting intuitive guess. Occipital gamma-augmentation enhanced during naming of unambiguous images can be explained by visual processing of stimuli with richer detail.

Significance: Our results support the theoretical model that guessing processes in visual domain occur following the accumulation of sensory evidence resulting from the bottom-up processing in the occipital-temporal visual pathways.

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1. Introduction

The ability to recognize a visually-presented object is included in the important brain functions to be preserved in epilepsy

surgery. Previous studies of patients with epilepsy using electrocorticography (ECoG) showed that naming of 'realistically-drawn or photographic objects' elicited augmentation of gamma activity at >50 Hz, sequentially involving bilateral occipital, bilateral occipital-temporal, left inferior-frontal and bilateral Rolandic regions (Cervenka et al., 2013; Kojima et al., 2013a). A number of studies using functional MRI (fMRI) have demonstrated hemodynamic activation in the aforementioned regions during naming of such unambiguous images (Graves et al., 2007; Garn et al., 2009;

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Liljeström et al., 2009). Electrophysiological and hemodynamic activation in the occipital and occipital-temporal regions is likely to reflect cortical processing involved in perception and attentive analysis of visual stimuli (Mishkin and Ungerleider, 1982; Goodale and Milner, 1992), while that in the left inferior-frontal and bilateral Rolandic regions involved in semantic generation of relevant answers and overt articulation (Cervenka et al., 2013; Kojima et al., 2013a).

The first question to be addressed in this ECoG study regards *what cortical regions* would be activated commonly and differentially during naming of ‘unambiguous’ and ‘ambiguous’ images. In our daily life, we sometimes need to judge what an object really is based on a vague and ambiguous silhouette. Recognition of such an image may require an intuitive guess rather than a meticulous analysis of the visual stimuli. Previous fMRI studies suggested that cortical processing for such a guess may be mediated at least partially by frontal or parietal lobes, based on the observations that tasks involving recognition of ambiguous images elicited greater hemodynamic activation in the orbitofrontal, inferior-frontal and inferior-parietal regions (Bar et al., 2006; Eger et al., 2007). Taking into account that hemodynamic activation on fMRI is tightly correlated with augmentation of gamma-band activity but not slower activities (Niessing et al., 2005), we hypothesize that naming of ‘ambiguous’ images, compared to that of ‘unambiguous’ ones, would elicit more intense gamma-augmentation in these frontal and parietal regions in the present study.

The second question regards *when* gamma-augmentation specific to the naming of ‘ambiguous’ images would take place. Analysis of averaged signals on magnetoencephalography (MEG) suggested that ‘a task required to recognize ambiguous images’ elicited cortical activation (presumably reflected by current dipole sources) in the occipital poles at 100 ms, in the left orbitofrontal region at 130 ms, and in the ventral occipital-temporal regions at 180 ms; thus, inferring that ‘top-down’ initial guess is exerted by the orbitofrontal region 50 ms earlier than ‘bottom-up’ systematic visual analysis in the occipital-temporal region (Bar et al., 2006). Taken together with the observations in the aforementioned neuroimaging and neurophysiology studies, we tested the hypothesis that gamma-augmentation specific to ‘ambiguous’ images would involve the frontal-parietal regions (including the orbitofrontal cortex) earlier than the occipital-temporal regions, bilaterally.

Measurement of ECoG signals provides a unique opportunity to externally validate previous observations in non-invasive neurophysiology studies. ECoG signals are directly recorded from the ventral and medial surfaces of cerebral cortex with a spatial resolution of 1 cm (Nagasawa et al., 2011; Uematsu et al., 2013). The signal-to-noise ratio is 20 to >100 times better on ECoG compared to scalp EEG recording (Ball et al., 2009). Conversely, it remains unclear if MEG truly detects discernible activities generated by deeply located cortices (Wennberg et al., 2011). Magnetic fields produced by cerebral cortex generally decrease as a function of $1/\text{distance}^2$ from the MEG sensors. The orbitofrontal gyrus is inherently >3 cm away from all MEG sensors. Furthermore, the risk of inaccurate estimation of deep sources in noninvasive recording has been suggested (Wang and Gotman, 2001), whereas ECoG studies do not require an additional analytic process to estimate the source of cortical activation. We determined whether our ECoG analysis can provide the data concordant with a theoretical model that rapid top-down guess selection in the orbitofrontal cortex precedes the bottom-up processing in the occipital-temporal visual pathway (Bar et al., 2006). Such rapid orbitofrontal activation justifying their top-down facilitation theory, to our best knowledge, has not been rigorously investigated for replication by other investigators.

2. Methods

2.1. Patients

This study has been approved by the Institutional Review Board at Wayne State University. The inclusion criteria consisted of (i) patients with focal epilepsy who underwent extraoperative subdural ECoG recording as a part of presurgical evaluation in Children’s Hospital of Michigan or Harper University Hospital in Detroit between December 2008 and November 2012; (ii) language mapping using measurement of gamma-augmentation elicited by naming of ‘unambiguous’ and ‘ambiguous’ images (Fig. 1); and (iii) written informed consent obtained by adult patients or the guardians of pediatric patients. The exclusion criterion consisted of structural lesion or seizure onset zone involving the occipital lobe. A total of 10 right-handed English-speaking patients satisfied the aforementioned criteria and were included in the present study (Table 1).

2.2. ECoG recording

Platinum macro-electrodes were surgically placed in the subdural space over the left, right, or bilateral cortical regions (intercontact distance: 10 mm; diameter: 4 mm; median: 112 electrode sites per patient [standard deviation: 23]). Placement of intracranial electrodes was clinically guided by the results of Phase-I presurgical evaluation, including scalp video-EEG recording, MRI, and 2-deoxy-2- ^{18}F fluoro-D-glucose (FDG) positron emission tomography (PET) (Asano et al., 2009). All electrode plates were stitched to adjacent plates or the edge of dura mater, to avoid movement of subdural electrodes after intracranial implantation. In all patients, intraoperative photographs were taken with a digital camera before dural closure as well as after re-opening during the second stage of surgery. All electrodes were displayed on the three-dimensional brain surface reconstructed from high-resolution MRI, as previously described in detail (Alkonyi et al., 2009; Wu et al., 2011). We confirmed the spatial accuracy of electrode display on the three-dimensional brain surface by using intraoperative digital photographs (Wellmer et al., 2002; Dalal et al., 2008).

ECoG signals were recorded for 3–5 days with a sampling rate of 1000 Hz, using a 192-channel Nihon Kohden Neurofax 1100A Digital System (Nihon Kohden America Inc, Foothill Ranch, CA, USA). The averaged voltage of ECoG signals derived from the fifth and sixth intracranial electrodes on the amplifier was used as the original reference; ECoG signals were then re-montaged to a common average reference (Korzeniewska et al., 2011). Channels contaminated with large interictal epileptiform discharges or artifacts were visually identified and excluded from the average, in order to minimize their contamination on ECoG traces. Usage of a common average reference is a widely-accepted practice in assessment of event-related gamma-augmentation recorded on subdural grid electrodes; its advantages and limitations were previously discussed (Crone et al., 2001; Nagasawa et al., 2011). Electrooculography (EOG) electrodes were placed 2.5 cm below and 2.5 cm lateral to the left and right outer canthi. ECoG traces were visually inspected with a band-pass filter with low-frequency cut-off at 53 Hz and sensitivity of 20 $\mu\text{V}/\text{mm}$; thereby, irregular broadband signals synchronized with facial and ocular muscle activities seen on EOG electrodes were treated as artifacts (Otsubo et al., 2008; Jerbi et al., 2009; Kovach et al., 2011; Kojima et al., 2013a). Seizure onset was defined as a sustained rhythmic change on ECoG accompanied by subsequent clinically typical seizure activity, not explained by state changes, and clearly distinguished from background ECoG and interictal activity (Asano et al., 2009). Electrode

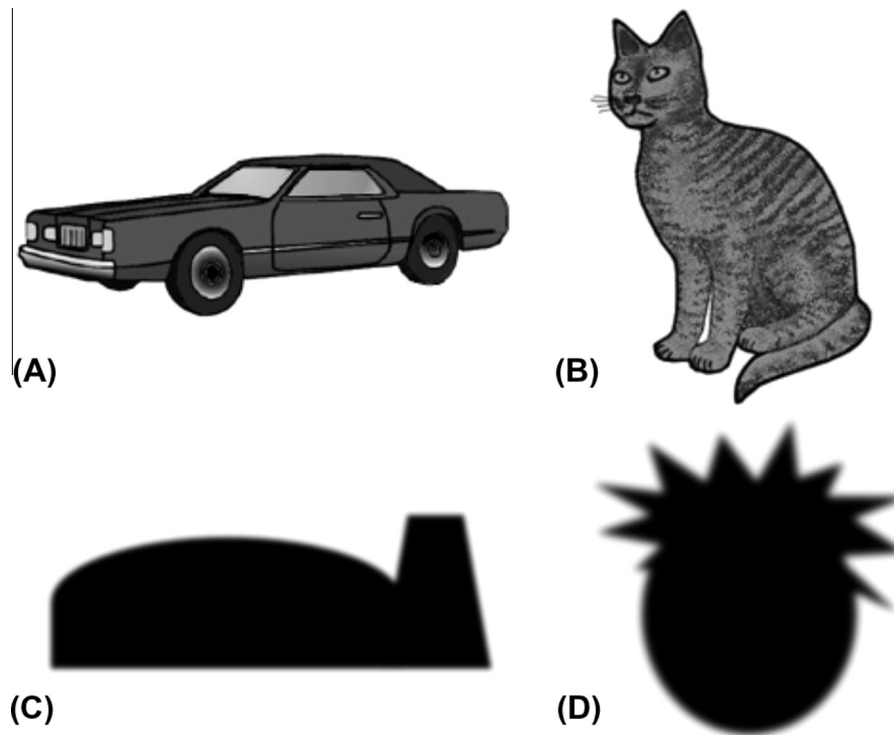


Fig. 1. Examples of visual stimuli. (A) and (B) were included in the unambiguous image set, while (C) and (D) in the ambiguous set. (C) was named, for example, as 'fish', 'whale' or 'submarine'. (D) as 'human', 'hair', or 'pineapple'.

Table 1
Patient profile.

Patient	Age at surgery	Handed-ness	Sampled regions	Seizure onset zone	AEDs	Neuropsychological data
#1	17	Rt	Lt F*PTO	Lt T	OXC, LEV	VCI: 109
#2	13	Rt	Rt FP	Rt Superior Rolandic	OXC	VCI: 106 CELLF: 118 PPVT: 128
#3	28	Rt	Rt F*PT	Rt T	CBZ	PPVT: 100.
#4	27	Rt	Lt F*PT	Lt F	LEV	PPVT: 91
#5	12	Rt	Lt F*PTO	Lt T	LTG, LCM	VCI: 98 CELLF: 109 PPVT: 95
#6	6	Lt	Lt F*PTO	Lt T	LTG, VPA	CELLF: 67 PPVT: 64
#7	15	Rt	Rt F*PTO	Rt T	LEV, ZNS, LTG	CELLF: 67 PPVT: 76
#8	17	Rt	Rt FPTO	Rt T	LEV, LCM	VCI: 107
#9	48	Rt	Lt FPTO	Lt T	LEV	VIQ: 72
#10	9	Rt	Lt F*PTO	Lt T	OXC	VCI: 91 CELLF: 84 PPVT: 85

Lt: Left. Rt: Right. F: Frontal. P: Parietal. T: Temporal. O: Occipital. *: Orbitofrontal cortex was sampled. OXC: Oxcarbazepine. LEV: Levetiracetam. CBZ: Carbamazepine. LTG: Lamotrigine. LCM: Lacosamide. VPA: Valproic acid. ZNS: Zonisamide. VCI: Verbal Comprehension Index. VIQ: Verbal IQ. CELLF: Clinical Evaluation of Language Fundamentals. PPVT: Peabody Picture Vocabulary Test. Patients #4–7 and #10: Female.

sites involved in seizure onset zone or structural lesions were excluded from further analysis (Jacobs et al., 2009; van Diessen et al., 2013).

2.3. Two naming tasks

Patients were comfortably seated on a bed in a dimly lit room, and instructed to overtly name each object as it is first perceived. Patients were instructed to overtly verbalize 'I don't know' when they were unable to provide an answer. Stimuli were presented sequentially on a 19-in. LCD monitor placed 60 cm in front of patients. Picture stimuli consisted of 60 realistically-drawn,

unambiguous images (Rossion and Pourtois, 2004) and 60 blurred silhouettes of ambiguous images (Fig. 1). Six patients were assigned naming of unambiguous images prior to that of ambiguous ones, while the remaining four patients were assigned naming of ambiguous ones first (two sessions about 10 min apart). Physical differences between unambiguous and ambiguous images include complexity and spatial frequency, which were higher in unambiguous ones. The size of picture stimuli ranged from 11 to 16 cm in height and width. Each stimulus was presented at the center of the monitor, in grayscale, on a black background, for 5000 ms with an inter-stimulus interval randomly ranging 2000–2500 ms. TTL trigger signals synchronized with the onset and offset of each

stimulus presentation were delivered to the ECoG recording system. These audible visual-language sessions were recorded using a Digital Voice Recorder (WS-300M, Olympus America Inc, Hauppauge, NY, USA) concurrently with ECoG recording, and the amplified audio waveform was integrated into the Digital ECoG Recording System (Brown et al., 2008). ECoG traces were aligned to: (i) stimulus (picture) onset; and (ii) response (answer) onset, as performed in our previous study (Kojima et al., 2013a). The response time was defined as the period between the onset of stimulus presentation and the onset of overt responses.

2.4. Time–frequency analysis

Each ECoG trial was transformed into the time–frequency domain using complex demodulation (Papp and Ktonas, 1977) via BESA software (BESA GmbH, Gräfelfing, Germany; Hoehstetter et al., 2004). A given ECoG signal was assigned an amplitude (a measure proportional to the square root of power) as a function of time and frequency (in steps of 10 ms and 5 Hz). The time–frequency transform was obtained by multiplication of the time-domain signal with a complex exponential, followed by a band-pass filter. The band-pass filter used here was a finite impulse response filter of Gaussian shape, making the complex demodulation effectively equivalent to a Gabor transform. The filter had a full width at half maximum of 2×15.8 ms in the temporal domain and 2×7.1 Hz in the frequency domain. The corresponding time–frequency resolution was ± 15.8 ms and ± 7.1 Hz (defined as the 50% power drop of the finite impulse response filter).

We determined ‘when’ and ‘where’ gamma activity at 50–120 Hz averaged across trials was augmented compared to the

preceding resting period (see the details in Kojima et al., 2013a). The time–frequency analysis relative to the onset of image presentation was designed to determine early gamma-augmentation time-locked to perception of images and particularly suitable to determine if the orbitofrontal region would induce rapid gamma-augmentation specific to ambiguous images. The analysis relative to the onset of responses was designed to determine late gamma-augmentation time-locked to generation of appropriate responses (Kojima et al., 2013a). We also determined whether the degree of such gamma-augmentation in each time–frequency bin reached significance using a studentized bootstrap statistic followed by Simes’ correction (Brown et al., 2008; Koga et al., 2011). Finally, sites surviving correction showing significant amplitude augmentation spanning (i) at least 20-Hz in width and (ii) at least 20-ms in duration were defined as significant gamma-augmentation elicited by a given task. We previously discussed the advantage and limitation of this analytic approach (Wu et al., 2011; Brown et al., 2012), which was, in short, validated by correlation with electrical stimulation data as well as post-surgical symptoms (Fukuda et al., 2008; Nagasawa et al., 2010a,b; Kojima et al., 2012; 2013a,b).

3. Results

3.1. Behavioral results

The mean response rate was 91.1% (median: 100%; standard deviation: 13.6%) in naming of ambiguous images and 100% in that of unambiguous ones. The response rate in naming of ambiguous images was somewhat smaller than that of unambiguous ones, though the difference did not reach significance ($p = 0.06$ on

Table 2
Location of gamma-augmentation elicited by naming tasks.

Side	Region	Number of sites showing significant gamma augmentation				Total number of analyzed sites
		Ambiguous images alone	Unambiguous images alone	Both	None	
Left	Superior-temporal	1 (2%)	3 (5%)	10 (18%)	42 (75%)	56
	Middle-temporal	0 (0%)	1 (3%)	2 (6%)	30 (91%)	33
	Inferior-temporal	2 (3%)	3 (5%)	4 (7%)	52 (85%)	61
	Medial-temporal	1 (11%)	1 (11%)	2 (22%)	5 (56%)	9
	Inferior-parietal	6 (11%)	2 (4%)	0 (0%)	45 (85%)	53
	Superior/middle-frontal	1 (3%)	0 (0%)	1 (3%)	33 (94%)	35
	Inferior-frontal	5 (10%)	2 (4%)	1 (2%)	43 (84%)	51
	Orbitofrontal	2 (5%)	0 (0%)	0 (0%)	36 (95%)	38
	Dorsolateral-premotor	3 (6%)	0 (0%)	13 (25%)	35 (69%)	51
	Medial-frontal	0 (0%)	2 (25%)	1 (13%)	5 (63%)	8
	Inferior-Rolandic	2 (4%)	2 (4%)	26 (51%)	21 (41%)	51
	Medial-occipital	0 (0%)	0 (0%)	6 (46%)	7 (54%)	13
	Polar-occipital	0 (0%)	2 (18%)	9 (82%)	0 (0%)	11
	Lateral-occipital	2 (11%)	2 (11%)	7 (37%)	8 (42%)	19
	Inferior-occipital-temporal	1 (2%)	6 (15%)	27 (66%)	7 (17%)	41
Right	Superior-temporal	0 (0%)	5 (14%)	3 (8%)	29 (78%)	37
	Middle-temporal	1 (4%)	0 (0%)	2 (9%)	20 (87%)	23
	Inferior-temporal	0 (0%)	0 (0%)	1 (6%)	17 (94%)	18
	Medial-temporal	0 (0%)	0 (0%)	0 (0%)	18 (100%)	18
	Inferior-parietal	0 (0%)	0 (0%)	1 (3%)	33 (97%)	34
	Superior/middle-frontal	1 (4%)	0 (0%)	1 (4%)	21 (91%)	23
	Inferior-frontal	4 (14%)	0 (0%)	3 (11%)	21 (75%)	28
	Orbitofrontal	1 (6%)	0 (0%)	0 (0%)	16 (94%)	17
	Dorsolateral-premotor	1 (4%)	1 (4%)	6 (21%)	20 (71%)	28
	Medial-frontal	0	0	0	0	0
	Inferior-Rolandic	1 (3%)	4 (13%)	10 (33%)	15 (50%)	30
	Medial-occipital	0 (0%)	0 (0%)	3 (43%)	4 (57%)	7
	Polar-occipital	0 (0%)	0 (0%)	3 (75%)	1 (25%)	4
	Lateral-occipital	1 (6%)	3 (19%)	3 (19%)	9 (56%)	16
	Inferior-occipital-temporal	0 (0%)	5 (36%)	5 (36%)	4 (29%)	14

In patient #4 who had the seizure onset zone involving the left frontal lobe, the left precentral gyrus showed gamma-augmentation during naming of both ambiguous and unambiguous images, while a left inferior frontal site showed gamma-augmentation specifically associated with ambiguous images. The present study failed to find significant difference in the location of sites showing gamma-augmentation across genders and ages.

Wilcoxon Signed Rank test). The response time in naming of ambiguous images was larger than that of unambiguous ones (median: 2385 vs 1265 ms; $p < 0.001$ on Wilcoxon Signed Rank test).

3.2. Spatial-temporal characteristics of naming-related gamma-augmentation

Table 2 summarizes the proportion of electrode sites showing significant gamma-augmentation elicited by naming of ambiguous and unambiguous images in each region of interest. In short, significant gamma-augmentation was noted in 161 out of the 530 sites (30.4%) in the left hemisphere and 69 out of the 297 (23.2%) in the right hemisphere. The left hemisphere was associated with a larger chance of showing significant gamma-augmentation compared to the right hemisphere ($p = 0.03$ on Chi-square test).

Within 200 ms from the onset of presentation of ambiguous or unambiguous images, gamma-augmentation sequentially and commonly involved the medial-occipital, polar-occipital, lateral-occipital and inferior-occipital-temporal regions, bilaterally (Figs. 2 and 3). Immediately prior to and during the responses during naming of both types of images, gamma-augmentation commonly

involved the dorsolateral-premotor and inferior-Rolandic regions, bilaterally (Fig. 2).

Twelve sites within inferior-frontal or orbitofrontal regions showed significant gamma-augmentation specific to ambiguous images, while only two in these regions unambiguous image-specific gamma-augmentation (Fig. 4). Six sites within the inferior-parietal region showed ambiguous image-specific gamma-augmentation, while only two sites of this region showed unambiguous image-specific gamma-augmentation (Fig. 4). None of these ambiguous image-specific gamma-augmentations took place within 400 ms following the onset of image presentation, but rather such gamma-augmentation was better time-locked to the onset of the response (Fig. 3).

In order to rule out the possibility that ambiguous image-specific gamma-augmentation in the orbitofrontal region was spuriously induced by ocular EMG artifacts (Jerbi et al., 2009; Nagasawa et al., 2011; Uematsu et al., 2013), we repeated the ECoG analysis on bipolar montage. Assessment on bipolar montage replicated orbitofrontal gamma-augmentation elicited by naming of ambiguous images (Fig. 5).

As an exploratory analysis, we determined whether the orbitofrontal regions would generate rapid event-related potentials (ERPs; Uematsu et al., 2013) following presentation of ambiguous

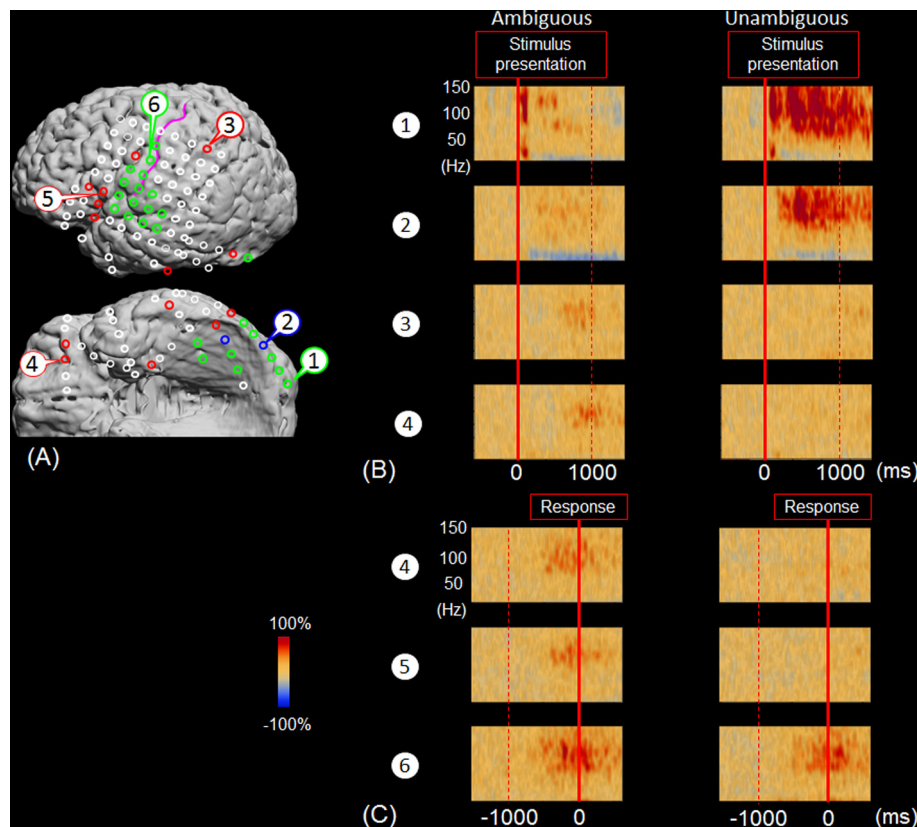


Fig. 2. Gamma-augmentation during naming tasks in patient #5. (A) Green electrodes: sites showing significant gamma-augmentation elicited by naming of unambiguous and ambiguous images. Blue electrodes: significant gamma-augmentation elicited by naming of unambiguous images alone. Red electrodes: significant gamma-augmentation elicited by naming of ambiguous images alone. White electrodes: no significant gamma-augmentation elicited by either naming task. (B) Time-frequency plots are presented with the zero point reflecting the onset of image presentation. Channel #1 in the left occipital pole showed significant gamma-augmentation within 100 ms following the onset of presentation of unambiguous and ambiguous images; naming of unambiguous images was associated with more sustained gamma-augmentation. Channel #2 in the lateral occipital region showed significant gamma-augmentation within 200 ms following the presentation of unambiguous images, while gamma-augmentation elicited by ambiguous images failed to reach significance. Channel #3 in the left inferior parietal lobule was associated with significant gamma-augmentation at 810–940 ms following the presentation of ambiguous images but no significant gamma-augmentation during naming of unambiguous ones. Channel #4 in the left orbitofrontal region was associated with significant gamma-augmentation at 860–1190 ms following the presentation of ambiguous images but no significant gamma-augmentation during naming of unambiguous ones. (C) Time-frequency plots are presented with the zero point reflecting the onset of responses. Channel #4 in the left orbitofrontal region and channel #5 in the left inferior-frontal region were associated with significant gamma-augmentation prior to responses to ambiguous images alone. Channel #6 in the left Rolandic region was associated with significant gamma-augmentation around the onset of responses to both ambiguous and unambiguous images. The mean response time was 1930 ms during naming of ambiguous images and 1140 ms during that of unambiguous ones. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

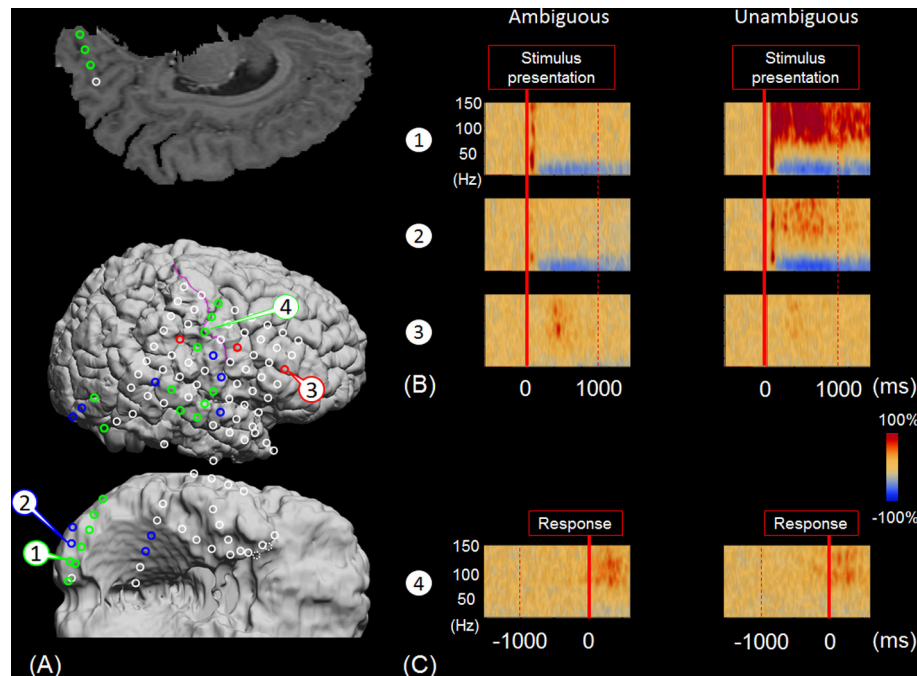


Fig. 3. Gamma-augmentation during naming tasks in patient #8. (A) Green electrodes: sites showing significant gamma-augmentation elicited by naming of unambiguous and ambiguous images. Blue electrodes: significant gamma-augmentation elicited by naming of unambiguous images alone. Red electrodes: significant gamma-augmentation elicited by naming of ambiguous images alone. White electrodes: no significant gamma-augmentation elicited by either naming task. (B) Time–frequency plots are presented with the zero point reflecting the onset of image presentation. Channel #1 in the right polar occipital region showed significant gamma-augmentation within 100 ms following the onset of presentation of unambiguous and ambiguous images; naming of unambiguous images was associated with more sustained gamma-augmentation. Channel #2 in the lateral occipital region showed significant gamma-augmentation within 200 ms following the presentation of unambiguous images, while gamma-augmentation elicited by ambiguous images failed to reach significance. Channel #3 in the right inferior frontal region was associated with significant gamma-augmentation at 410–530 ms following the presentation of ambiguous images, but gamma-augmentation during naming of unambiguous ones failed to reach significance. (C) Time–frequency plots are presented with the zero point reflecting the onset of responses. Channel #4 in the right Rolandic region was associated with significant gamma-augmentation around the onset of responses for both ambiguous and unambiguous images. The mean response time was 2580 ms during naming of ambiguous images and 1750 ms during that of unambiguous ones. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

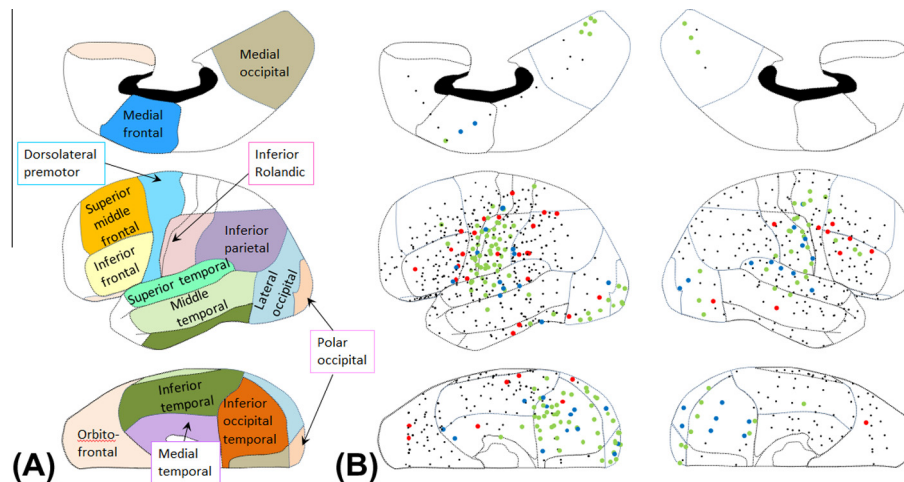


Fig. 4. Gamma-augmentation elicited by naming of ambiguous and unambiguous images. (A) Presented is the definition of the anatomical regions of interest; this was employed in our previous study (Matsuzaki et al., 2012; Kojima et al., 2013a). Medial occipital region: medial portion of BA 17/18. Polar occipital region: polar portion of BA 17/18. Lateral occipital region: lateral portion of BA 19/37. Inferior occipital-temporal region: inferior portion of BA 19/37. Medial temporal region: BA 27/28/34/35/36. Inferior temporal region: inferior temporal gyrus involving BA 20/37. Middle temporal region: middle temporal gyrus involving BA 21/37. Superior temporal region: BA 22/41/42. Inferior parietal region: BA 39/40. Inferior Rolandic region: BA 4/3/1/2 not more than 4 cm superior from the sylvian fissure. Medial frontal region: medial portion of BA 6/8 and posterior portion of BA 24/32/33. Dorsolateral premotor region: dorsolateral portion of BA 6. Middle/superior frontal region: lateral portion of BA 46/9/8. Inferior frontal region: inferior frontal gyrus involving BA 44/45. Orbitofrontal region: BA 47/11. (B) Red dots: electrode sites showing significant gamma-augmentation elicited by naming of ambiguous images alone. Blue dots: sites showing significant gamma-augmentation elicited by naming of unambiguous images alone. Green dots: sites showing significant gamma-augmentation elicited by both naming tasks. Black dots: sites failing to show significant gamma-augmentation during either naming task. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

images, while we found ambiguous image-specific gamma-augmentation taking place at least 400 ms after the onset of image presentation. We performed this exploratory analysis, because

rapid cortical activation in the orbitofrontal region was previously inferred by the presence of rapid event-related magnetic fields on averaged MEG traces (Bar et al., 2006). We evaluated ERPs with a

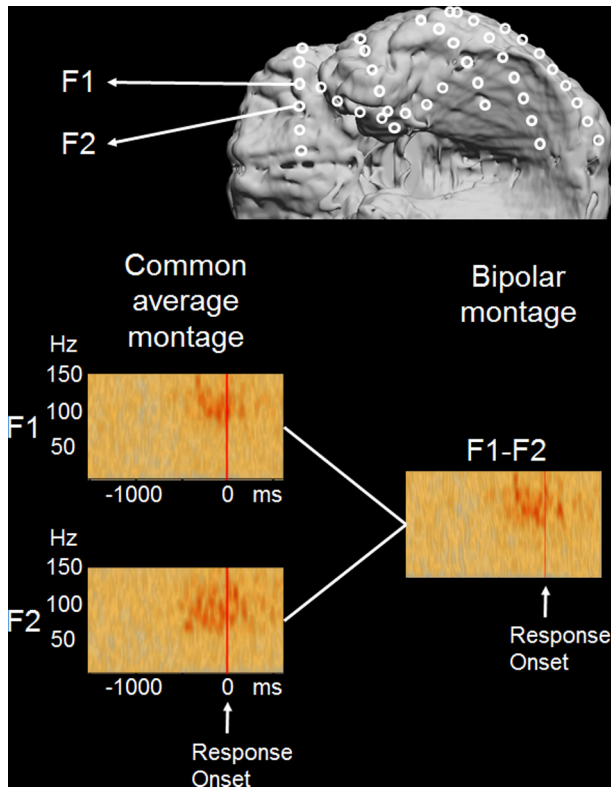


Fig. 5. Gamma-augmentation specifically elicited by naming of ambiguous images. Both on common average and bipolar montages, electrodes F1 and F2 showed significant gamma-augmentation immediately prior to the responses during naming of ambiguous images. It has been reported that bipolar montage can effectively eliminate ocular EMG-elicited gamma-augmentations (Jerbi et al., 2009; Nagasawa et al., 2011; Uematsu et al., 2013).

high-pass filter of 0.1 Hz and a low-pass filter of 20 Hz, as applied in Bar et al., 2006. In short, on bipolar montage, we failed to observe rapid ERPs in the orbitofrontal region preceding those in the occipital-temporal regions (Supplementary Fig. S1).

Four sites within the medial-, polar-, lateral-occipital regions as well as inferior-occipital-temporal region showed gamma-augmentation specific to ambiguous images, while eighteen in these regions showed unambiguous image-specific gamma-augmentation (Fig. 4). The 2×3 Fisher's exact probability test revealed that the proportion of sites showing ambiguous image-specific gamma-augmentation was higher in the 'frontal-parietal network' including the inferior-frontal/orbitofrontal and the inferior-parietal regions, compared to the 'occipital-temporal network'. In contrast, the proportion of sites showing unambiguous image-specific gamma-augmentation was higher in the 'occipital-temporal network' compared to the aforementioned 'frontal-parietal network' ($p < 0.0001$; Table 2).

4. Discussion

4.1. Spatial characteristics of gamma-augmentation specific to naming of ambiguous images

The present study using ECoG recording externally validated the previous neuroimaging observations that tasks involving visual recognition of ambiguous images elicited greater cortical activation in the frontal-parietal networks (including orbitofrontal, inferior-frontal and inferior-parietal regions), bilaterally (Bar et al., 2006; Eger et al., 2007). Taking into account that the response time was consistently longer during naming of ambiguous images

compared to that of unambiguous ones, such frontal-parietal gamma-augmentation specific to ambiguous images may reflect the additional cortical processing involved in exerting intuitive guess. 'Guess' is a short single word, but what it describes requires complex processes including 'estimation', 'semantic analysis', and 'selection'. Behavioral assessments in the present study and another have shown that a visual recognition task involving guess resulted in longer response times by about a second (Bar and Ullman, 1996). It is difficult to clearly segregate the functional significance of ambiguity-recognition-related gamma-augmentation in each frontal-parietal site, solely based on the observations in the present study; for example, we did not design a task specifically posing 'selection' without 'semantic analysis'. Furthermore, 'estimation' of common objects would inherently involve 'semantic analysis'.

A number of fMRI studies have reported that tasks requiring 'estimation' and 'selection' in the visual domain were associated with hemodynamic activation in the inferior-frontal, orbitofrontal, and inferior-parietal regions, bilaterally (Heekeren et al., 2004; Bar et al., 2006; Summerfield et al., 2006; Eger et al., 2007; Thielscher and Pessoa, 2007). A task requiring 'estimation' and 'selection' but presumably not 'semantic analysis' also elicited hemodynamic activation in these regions (Grinband et al., 2006). In an fMRI study (Ploran et al., 2007), a picture of a common object was initially masked and gradually revealed to participants every 2 s; thereby, hemodynamic activation in the occipital regions was correlated to the richness of stimulus information, whereas that in the frontal, parietal and temporal regions built up and peaked at the time of recognition of a given object (Ploran et al., 2007). In another fMRI study of 'estimation' and 'selection' in the auditory domain reported a linear association between a longer response time and a larger hemodynamic activation in the inferior-frontal regions, bilaterally (Binder et al., 2004). Furthermore, lesions of the inferior-frontal and orbitofrontal regions are generally associated with impairment in decision making in the visual domain (Clark et al., 2003; Tsuchida and Fellows, 2012) as well as impairment in semantic processing (Hart and Gordon, 1990; Jefferies and Lambon Ralph, 2006). The observations in these imaging and lesion studies are consistent with the notion that frontal-parietal gamma-augmentation specific to ambiguous images reflects cortical processing involved in either 'estimation', 'semantic analysis', and/or 'selection'.

4.2. Temporal characteristics of frontal-parietal gamma-augmentation specific to naming of ambiguous images

The present ECoG study demonstrated that frontal-parietal gamma-augmentation specific to ambiguous images took place at least 400 ms after the onset of image presentation, while gamma-augmentation in the occipital and occipital-temporal regions consistently occurred within 200 ms. Our ECoG observations support the theoretical model that guessing processes in the visual domain take place following the accumulation of sensory evidence resulting from the bottom-up processing in the occipital-temporal visual pathways (Ploran et al., 2007; Heekeren et al., 2008; Fig. 6). Conversely, the present study failed to provide the data supporting another model proposing that rapid top-down guess process in the orbitofrontal region takes place about 130 ms after the onset of image presentation and precedes the bottom-up processing in the occipital-temporal regions (Bar et al., 2006; Kveraga et al., 2007; Fig. 6).

The present study provided robust evidence that ambiguous-image-related cortical activation took place in portions of frontal-parietal regions at >400 ms after the onset of image presentation. Such late orbitofrontal gamma-augmentation was replicated both on bipolar and common average montages (Fig. 5). It should be

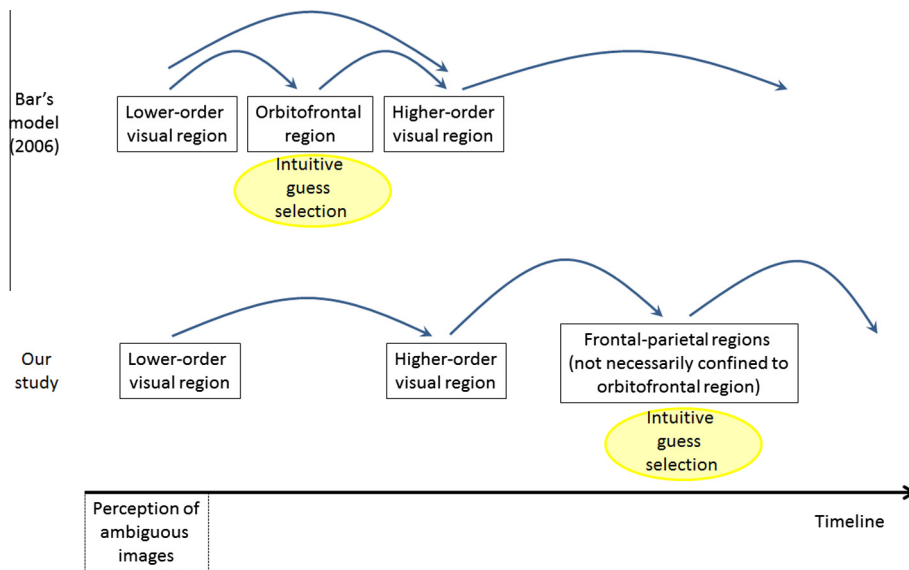


Fig. 6. Models of intuitive guess. A theoretical model (upper) proposes that the lower-order visual region rapidly projects the information of blurred images to the orbitofrontal region possibly via the magnocellular pathway, and deliver a top-down signals back to the occipital-temporal region including the fusiform gyrus to facilitate visual recognition (Bar et al., 2006; Kveraga et al., 2007). The present study supports the model (lower) that guessing processes in the visual domain take place in the frontal-parietal regions (not necessarily confined to the orbitofrontal regions) following the accumulation of sensory evidence resulting from the bottom-up processing between the lower- and higher-order visual pathways (Ploran et al., 2007; Heekeren et al., 2008).

noted that humans unconsciously repeat miniature saccades and fixation in daily life. Such saccades, which may be better detected by the eye tracking system, most frequently take place around 100–300 ms following the presentation of picture stimuli (Fletcher-Watson et al., 2008; Crouzet et al., 2010). A number of studies have reported that each saccade generates artifact signals in the peri-orbital regions on scalp EEG (Yuval-Greenberg et al., 2008), on MEG (Carl et al., 2012) and even on intracranial ECoG (Jerbi et al., 2009; Nagasawa et al., 2011; Uematsu et al., 2013). In the present study, re-assessment of event-related gamma activity was employed on bipolar montage (Jerbi et al., 2009). We previously demonstrated that peri-orbital (ocular-EMG-oriented) gamma-augmentation on common average montage exactly time-locked to saccades was effectively eliminated on bipolar montage, while occipital gamma-augmentation of cortical origin remained robust on both common average and bipolar montages (Nagasawa et al., 2011; Uematsu et al., 2013). The benefit of bipolar montage is to eliminate contamination by distant ocular activities (Worrell et al., 2012), whereas the drawback is an inability to directly discriminate which electrode within a pair indeed gives rise activities of interest. We used bipolar montage as a tool complementary to our single electrode analysis.

Event-related augmentation of gamma activity is generally considered an excellent summary measure of local cortical activation (Lachaux et al., 2012), since it is tightly correlated to the firing rate on single neuron recording (Ray et al., 2008) and hemodynamic activation on fMRI (Niessing et al., 2005; Scheeringa et al., 2011). Furthermore, a study of the occipital lobe of monkeys showed that cortical sites showing event-related gamma-attenuation were associated with decreased firing rates and hemodynamic deactivation (Shmuel et al., 2006). Electrical stimulation of sites showing event-related gamma-augmentation frequently elicits concordant sensory, motor, or language symptoms (Miller et al., 2007; Fukuda et al., 2008; Nagasawa et al., 2010a,b; Kojima et al., 2012). Furthermore, surgical resection of sites showing event-related gamma-augmentation frequently resulted in functional impairment (Cervenka et al., 2013; Kojima et al., 2013a,b). Measurement of event-related gamma activity, compared to alpha/beta

activity, is better capable of assessing the rapid dynamism of cortical activity, partly because a single oscillation cycle is 20 ms for gamma activity at 50 Hz and 50 ms for beta activity at 20 Hz. Moreover, cortical activation cannot be uniformly reflected by attenuation or augmentation of alpha/beta activities (Niessing et al., 2005; Crone et al., 2006; Fukuda et al., 2010; Scheeringa et al., 2011).

The present ECoG study does not disprove the validity of the observations reported by Bar et al. (2006), but simply failed to provide evidence justifying the presence of orbitofrontal cortical activation as rapid as 130 ms. First of all, MEG was not simultaneously recorded with ECoG in the present study, and the assigned tasks differed between two studies. We cannot rule out the possibility that subdural electrodes may have failed to sample potential generators of rapid cortical activation, while ECoG signals were sampled from 55 sites in the orbitofrontal regions (Fig. 4) not involved in seizure onset zones or epileptogenic lesions. It is still possible that rapid top-down facilitation is exerted by orbitofrontal sites different from those showing late gamma-augmentation (Fig. 2).

In the present study, ambiguous images elicited rapid positive ERPs initially in the left occipital pole and subsequently in the left inferior occipital-temporal region within 150 ms; these ERPs were replicated on bipolar montage (Supplementary Fig. S1). The amplitudes of such ERPs in the left inferior occipital-temporal region (within the vicinity of the fusiform gyrus) were about 20 to >100 times larger than the amplitude of N170 previously reported in non-invasive electrophysiology studies (Itier and Taylor, 2004; Zion-Golumbic and Bentin, 2007; Sadeh and Yovel, 2010). Conversely, the orbitofrontal sites (showing late gamma-augmentation on time-frequency analysis; Fig. 2) demonstrated positive deflections on common average montage at 200 ms, but such rapid orbitofrontal signal deflections failed to be replicated on bipolar montage (Supplementary Fig. S1). Thus, such rapid signal deflections may not be generated by the orbitofrontal cortex locally. A left EOG electrode placed 2.5 cm below and 2.5 cm lateral to the left outer canthus showed a small signal deflection (Supplementary Fig. S1). A previous human study using cortico-cortical evoked

potentials suggested cortico-cortical propagation to be as rapid as 40 ms per 10 cm (Matsumoto et al., 2012), whereas the distance between the lower-order visual cortex and the orbitofrontal cortex is about 15 cm (Thiebaut de Schotten et al., 2012). Taken together, we failed to collect external evidence supporting that the orbitofrontal cortex receives 'macroelectrode-detectable' signals triggered by the lower-order visual cortex as rapidly as 130 ms following the image presentation.

4.3. Significance of gamma-augmentation specific to unambiguous images

Neural responses in the occipital and occipital-temporal regions are generally believed to depend on the physical properties of visual stimuli, whereas the inferior-frontal, orbitofrontal, or inferior-parietal regions are not generally considered as specifically responding to such physical properties. Unambiguous images used in the present study had higher complexities and spatial frequencies compared to ambiguous images (Fig. 1). A number of previous studies reported that visual stimuli with richer detail were associated with greater hemodynamic activation and ECoG gamma augmentation in the occipital and occipital-temporal regions, bilaterally (Ploran et al., 2007; Vidal et al., 2010; Engell and McCarthy, 2011). Another ECoG study reported that increased attention to visual stimuli also contributed to greater gamma augmentation (Engell and McCarthy, 2010). Our previous ECoG study demonstrated that saccades differentially elicited gamma-modulations in the occipital regions (Uematsu et al., 2013); the polar and lateral occipital regions showed large saccade-related gamma-attenuation followed by gamma-augmentation at the offset of saccades, whereas the medial-occipital and inferior-occipital-temporal regions showed small saccade-related gamma-attenuation followed by large gamma-augmentation at the offset of saccades. Further studies of visual recognition using eye tracking systems are warranted, in order to determine how much of the cortical activation on imaging and electrophysiological data can be attributed to eye movements during a given task.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.clinph.2014.03.034>.

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